

Research Statement

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My research goal is to enable robots to assist humans in various daily tasks intelligently and reliably. To achieve this goal, my previous research covers three topics: 1) applying foundation models for the preference feedback in reinforcement learning, 2) modular robot learning for adaptation in novel environments, and 3) adversarial learning for robust control.

Robot Learning with Foundation Model. Conventional robot learning approaches to achieve complex robotic control tasks via reward engineering, motion capture data, human demonstrations, or human preference annotations. Among them, reinforcement learning from human feedback (RLHF) leverages human preferences to train predictors for predicting preference rewards and achieves promising results on teaching robots versatile behaviors. However, this method requires extensive human labor for providing the preference feedback on numerous pairs of rollout trajectory videos.

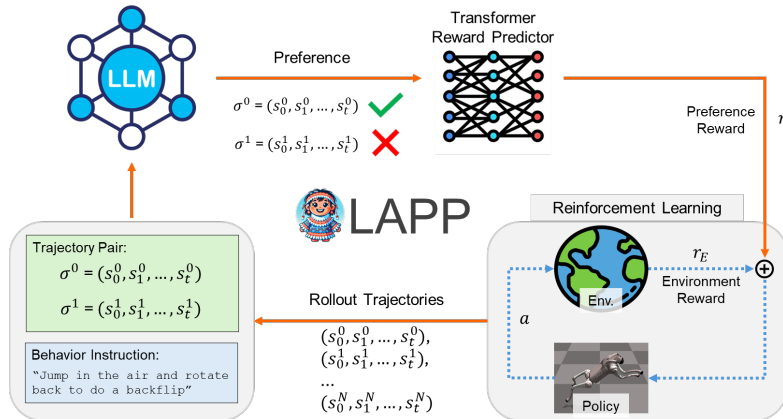


Figure 1: **LLM-Assisted Preference Prediction (LAPP)** [1]. LAPP generates preference feedback from an LLM based on rollout trajectories pairs of raw state and actions as well as a high-level behavior instruction. A transformer-based reward predictor is trained using these preferences while simultaneously optimizing a robot policy to maximize a weighted sum of environment rewards and predicted preference rewards.

We aim to leverage the strong reasoning capability of LLMs to automate this human preference annotation process and propose the LLM-Assisted Preference Prediction (LAPP) [1]. LAPP prompts LLMs to generate preference feedback from state-action trajectories and uses it to guide the reinforcement learning (RL) process. It enables robots to learn efficient, customizable, and yet challenging behaviors with minimum human efforts.

To the best of my knowledge, LAPP is the first work that verifies the feasibility to generate preference feedback from LLMs directly on raw state-action trajectories and accomplish challenging tasks that cannot be solved by explicit single reward function engineering, such as the quadruped backflip. In the future research, I would like to study more thoroughly how the noisy outputs from the LLMs affect the stability of the performance of LAPP, and further iterate this pipeline to be more robust against potential errors in the LLMs outputs. Moreover, with the continuous improving of the reasoning capability of the vision language models (VLMs), I am interested in extending the idea of LAPP to rollouts with image states and considering the tasks that requires visual inputs to the policy networks such as reactive motion planning in robot locomotion.

Adaptation to novel environments. Robots are typically trained to excel at specific tasks such as relocating objects or navigating to predetermined locations. However, such robots need to be re-trained from scratch when faced with new tasks, body changes, or visual configuration changes. This retraining process is usually time-consuming and computationally expensive, hindering the deployment of robots in the real world where the environment is changing continuously and sometimes drastically. In contrast, humans demonstrate remarkable capabilities to continuously acquire new skills by drawing on past experiences. Even under physical constraints imposed by injuries, we can still rapidly adapt to perform new tasks. Despite significant advancements in robotics and machine learning, robots still cannot generalize their experiences across a wide range of tasks, body configurations, or visual settings. It is crucial to empower robots with an adaptation capability similar to that of humans if we want to deploy them in the real world.

Previous works have attempted to enable fast adaptation to novel environments by regularizing a large policy network for multi-task learning. They achieve this by learning routing connections to reuse part of the network weights [2] or by assigning task-specific sub-networks [3]. However, the capacity of the large policy network grows exponentially as the number of tasks increases, leading to enormous memory consumption and computational expense.

Motivated by the need for compact yet highly adaptable robot policies, I adopted a **modular structure design** of the policy networks and developed the Policy Stitching algorithm [4] and the Perception Stitching algorithm [5]. The key idea behind these two methods is to disentangle the knowledge of different objectives such as tasks, sensors, robot kinematics, and then learn them in different neural network modules. This allows us to reuse modules trained in different environments by assembling them into novel combinations and deploying them in previously unseen environments. For instance, given one policy trained for a 3-DoF manipulator to pick up a cube and another policy trained for a 5-DoF manipulator to pick up a stick, if we would like to have the 3-DoF manipulator pick up a stick now, Policy Stitching algorithm [4] can directly take the robot module in the first policy and stitch it with the task module in the second policy.

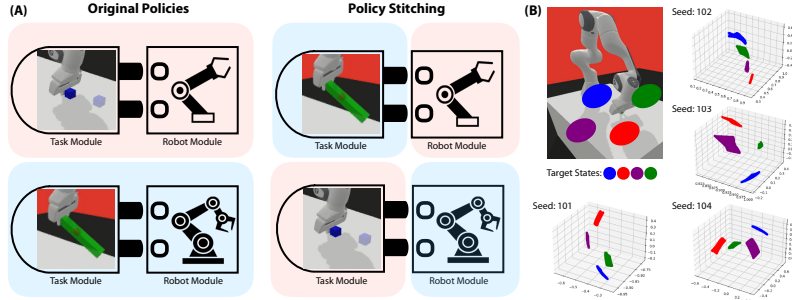


Figure 2: **Policy Stitching** [4]. (A) Our framework facilitates robot transfer learning among novel combinations of robots and tasks by decoupling and stitching robot and task modules. (B) Motivation example: A robot arm is trained to reach goals in four different target regions using the modular policy. Results from separate training runs with different random seeds (101-104) show misaligned latent representations.

Through my work on **training neural network modules to be reusable** in other environments, I have found that the latent representation misalignment issue [6] between the encoders and decoders trained in different environments is the major problem that diminishes the performance of the reassembled policy in a new environment. This observation drives my research into developing techniques inspired by the relative representation algorithm [6] and the covariance regularization method [7, 8, 9, 10] to enforce the invariance at the latent space of different encoders. Our experience has shown that training the latent states to be invariant to isometric transformations and enforcing the latent features to be independent can significantly alleviate the latent space misalignment issue and improve the performance of the reassembled policy. Although proven to be successful in many real-world experiments, my current approaches cannot achieve very satisfying performance on long-horizon dexterous tasks yet, leading to my future research to further understand the latent space of neural networks for standardizing the interfaces between different network modules.

Robustness against disturbances. Many robot learning approaches show satisfactory performance in the ideal disturbance-free environments. However, in a real-world environment, noise and disturbances are inevitable, which often lead to a drastic drop in robot performance. If we want to deploy robots in the real world, we need control policies that are not only robust to external disturbances caused by outside perturbations, but also robust to internal disturbances caused by hardware noise and errors. To this end, previous works [11, 12] have proposed various data augmentation methods that inject random noise into the trajectory data. However, these methods generally cannot be effective against more malicious disturbances beyond random noise. For example, suppose a protagonist robot arm wants to pick up a cube on the desk, while there is another adversarial robot arm deliberately blocks in the way of the protagonist or moves the cube away from the grasp of the protagonist. In this case, a robot trained with data augmentation techniques can no longer be robust enough to accomplish the task. How can we teach a robot to avoid attacks and recover from failures in an adversarial environment?

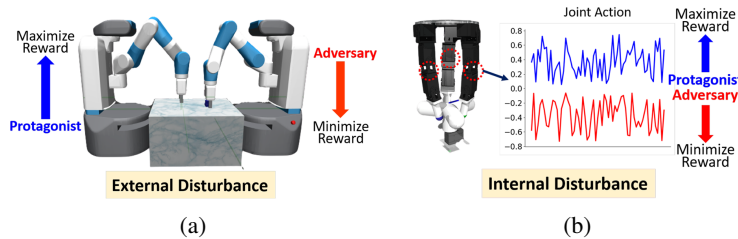


Figure 3: **Adversarial skill learning for robust manipulation** [13]. The protagonist tries to maximize the reward while the adversary tries to minimize it. (a) An external disturbance, the protagonist is disturbed by another adversary robot. (b) An internal disturbance, the protagonist joint action is under the disturbance of adversarial action on each joint of the robot.

Inspired by the training pipeline of the Generative Adversarial Network (GAN) [14], I proposed the Adversarial Skill Learning method [13]. The key contribution of this approach lies in the **adversarial pipeline design** of the training process. Specifically, we introduce an adversary agent to disturb the protagonist agent. During the reinforcement learning process, the protagonist agent is optimized to accomplish the task, while the adversary agent is optimized to obstruct the task. These two phases of optimization occur iteratively. After this competitive GAN-style training process, the adversary agent learns dexterous skills to attack the protagonist agent, while the protagonist agent is trained to be robust against malicious adversarial attacks as well as various random perturbations.

While my proposed method shows a pronounced advantage over vanilla data augmentation, it still requires manually tuning the optimal amplitude range of the disturbance for the training. In my future research, I aim to enhance the Adversarial Skill Learning framework with the recent advances in the diffusion policy [15] and the foundation models [16]. Along this path, I believe I can push the limits of robot control robustness, enabling reliable deployment of robots in the real world.

Looking further, my long-term goal is to develop general embodied intelligence that assists people in every aspect of their work and daily lives. In the era of foundation models, we can leverage the continuously growing knowledge and intelligence of them to push the limits of robustness and adaptation capabilities of robots that I previously focused on, as well as many other performance facets that I am excited to explore.

References

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